Experiment 13

Experiment to handle missing values and inappropriate data in a given data set

Aim:

To handle missing values and inappropriate data in a given data set

Code:

```
import numpy as np
import pandas as pd

df = pd.read_csv("Hotel_Dataset.csv")
df

df.duplicated()
df.info()

df.drop_duplicates(inplace=True)
df

len(df)

index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
df

df.drop(['Age_Group.1'], axis=1, inplace=True)
df

df.CustomerID.loc[df.CustomerID < 0] = np.nan
df.Bill.loc[df.Bill < 0] = np.nan
df.EstimatedSalary.loc[df.EstimatedSalary < 0] = np.nan
df

df['Rating(1-5)'].loc[(df['Rating(1-5)'] < 1) | (df['Rating(1-5)'] > 5)] = np.nan
df['NoOfPax'].loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20)] = np.nan
df

df.Age_Group.unique()
df.Hotel.unique()

df.Hotel.replace(['Ibys'], 'Ibis', inplace=True)

df.FoodPreference.unique()
df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg', inplace=True)
df.FoodPreference.replace(['non-Veg'], 'Non-Veg', inplace=True)

df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()), inplace=True)
df.NoOfPax.fillna(round(df.NoOfPax.median()), inplace=True)
df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
df.Bill.fillna(round(df.Bill.mean()), inplace=True)
df
```

Output:

```
In [1]: df = pd.read_csv("Hotel_Dataset.csv")
In [2]: df.duplicated()
Out[2]:
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool

In [3]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     object
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     object
 4   FoodPreference   11 non-null     object
 5   Bill             11 non-null     int64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group.1      11 non-null     object

In [4]: df.drop_duplicates(inplace=True)
In [5]: index = np.array(list(range(0, len(df))))
In [6]: df.set_index(index, inplace=True)
In [7]: df.drop(['Age_Group.1'], axis=1, inplace=True)
In [8]: df
Out[8]:
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000
3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000
5	6	35+	3	Ibys	Non-Veg	1909	2	122220
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122
7	8	20-25	7	LemonTree	Veg	2999	-10	345673
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999
9	10	30-35	5	RedFox	non-Veg	6755	4	87777

```
In [9]: df.CustomerID.loc[df.CustomerID < 0] = np.nan
In [10]: df.Bill.loc[df.Bill < 0] = np.nan
In [11]: df.EstimatedSalary.loc[df.EstimatedSalary < 0] = np.nan
In [12]: df['Rating(1-5)'].loc[(df['Rating(1-5)'] < 1) | (df['Rating(1-5)'] > 5)] = np.nan
In [13]: df['NoOfPax'].loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20)] = np.nan

In [14]: df.Hotel.replace(['Ibys'], 'Ibis', inplace=True)
In [15]: df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg', inplace=True)
In [16]: df.FoodPreference.replace(['non-Veg'], 'Non-Veg', inplace=True)

In [17]: df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()), inplace=True)
In [18]: df.NoOfPax.fillna(round(df.NoOfPax.median()), inplace=True)
In [19]: df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
In [20]: df.Bill.fillna(round(df.Bill.mean()), inplace=True)
In [21]: df
Out[21]:
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	Veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	4.0	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	4.0	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-25	4.0	LemonTree	Veg	2999.0	2.0	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5.0	RedFox	Non-Veg	6755.0	4.0	87777.0

Result:

Missing and inappropriate data in the hotel booking dataset were successfully handled. Duplicate records were removed, invalid negative values were replaced with NaN, and missing entries were imputed,

Result:

Missing and inappropriate data in the hotel booking dataset were successfully handled. Duplicate records were removed, invalid negative values were replaced with NaN, and missing entries were imputed, resulting in a clean and consistent dataset ready for analysis.